



DATA STRUCTURES II

COMP 272 & COMP 400C— Spring 2026

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The policies laid out in this syllabus are subject to change due to unforeseen circumstances, scheduling conflicts, etc. Any changes will be announced in class and posted online.

Catalog Description

This course studies advanced abstract data types, such as sets, maps, and graphs, and their implementation using arrays and dynamically allocated nodes in an object-oriented language. The course also studies the performance of the data structures built-in operations and related algorithms such as sorting, searching, and traversing.

Learning Objectives

By the end of this course, students will :

- understand the key properties (structure, methods, and performance of methods) of several data structures (primarily non-linear ones) including priority queues, trees, graphs, and hash tables;
- be able to implement and use sorting, searching and traversing algorithms;
- be able to solve computational problems by designing suitable algorithms and integrating efficient data structures;
- understand how to analyze and compare the efficiency of algorithms.

Textbooks

Data Structures and Algorithm Analysis in Java, 3rd Edition, by Mark Allen Weiss (Pearson, 2021).

Please note that the textbook is not required explicitly (i.e. assignments and other class exercises do not rely on the student purchasing the textbooks). However, it is highly recommended that students use the textbook as a guide to keep up with weekly reading, and as a source of examples.

Prerequisites

COMP 271, (COMP 163 or MATH 201) and (MATH 131 or MATH 161).

Course Modality

Lectures are delivered in-person and on-campus (Lake Shore Campus) every Monday, Wednesday, and Friday. The meeting times and locations per section are listed in Table 1. Occasionally,

certain events (extreme weather, scheduling conflicts, etc.) may require lectures to be delivered via Zoom (accessed from SAKAI) either synchronously or asynchronously. Changes to course meetings, deadlines, and quiz dates **will always be** communicated in class and via SAKAI announcements. Students are expected to periodically check their email and announcements.

Section	Days	Time	Location
001	Mon/Wed/Fri	12:35 PM – 1:25 PM	Mundelein Center, Room 304
002	Mon/Wed/Fri	1:40 PM – 2:30 PM	Mundelein Center, Room 608

Table 1: Lecture meeting times and locations by section

Assignments and Quizzes

Assignment specifications are uploaded to the Sakai platform and completed assignments are submitted via Sakai (email submissions are not acceptable). Programs that do not successfully compile without errors will usually receive a failing grade.

Each student will be allowed one late assignment of their choice. Think of this as having one “Late OK” pass. Late Passes allow you to turn in an assignment up to 24 hours after the due date. Once your this Late Pass is used, no more assignments will be accepted after the due date. There are no exceptions except for documented emergencies.

Grading

The course grade is computed with the following weights:

- Assignments & Quizzes: 30%
- Midterm Exam: 30%
- Final Exam: 30%
- Attendance & Participation: 10%

In order to be eligible for a passing grade in the course, at least 2 of the programming assignments must be submitted in good faith (i.e. substantial solution progress made and no associated academic integrity issues). For example, if a student’s final course average is an 80 %, but only 1 programming assignment is submitted, the student will receive a D in the course, as opposed to a B. Eligibility to pass does not guarantee a passing grade.

Grading Scale:

≥ 93	A	82 - 86	B	68 - 73	C	50 - 55	D
90 - 92	A-	79 - 81	B-	62 - 67	C-	< 50	F
87 - 89	B+	74 - 78	C+	56 - 61	D+	-	-

Tutoring

You have the option of meeting with tutors (at LSC) for help with assignments. Please refer to the following link for tutoring options: CS Department Tutoring hours.

Note that certain tutors may not have proficiency in the material of the course, so please confirm with the tutor prior to scheduling an appointing/dropping-in.

Office Hours

Office hours by appointment via calendly: <https://calendly.com/kennaoui-luc> , in *Doyle Center Room 304* or via Zoom.

Exams

The course will have an in-person midterm exam and a final exam. In-person attendance is required. The official dates for our final exams are given in LOCUS.

Attendance

Attendance during lectures is expected and a portion of the final grade is associated with attendance/participation. Furthermore regular attendance/participation is correlated with higher grades in the course. Students who arrive in the first 10 minutes or so of class are generally treated as attending the entire class.

If you need to miss class for a religious holiday, please alert me in advance so that will not be counted as an absence. As needed, conflicting due dates for quizzes and assignments can be adjusted.

Accommodations

Accommodations for coursework must be approved by the Student Accessibility Center (SAC), and a Letter of Accommodation (LOA) must be provided to the Instructor at the start of the semester. Requests for accommodations will not be granted without a LOA.

Student Accessibility Center (SAC)

When applicable, it is the **responsibility of the student** to schedule exams with the Student Accessibility Center (SAC) **seven days in advance** of the exam/quiz. Students that fail to do this may take their quiz during the regularly scheduled class time (in class), but may not receive the requisite timing and/or distraction free environment that is offered by SAC. No makeup quizzes are administered.

Academic Integrity

Students are expected to have read the statement on academic integrity available at https://www.luc.edu/academics/catalog/undergrad/reg_academicintegrity.shtml.

This policy applies to the course. The minimum penalty for academic dishonesty is a grade of F for that assignment. Multiple instances or a single severe instance on a major exam or assignment

may result in a grade of F for the course. All cases of academic dishonesty will be reported to the department office and the relevant college office where they will be placed in your school record.

Academic dishonesty includes, but is not limited to, working together on assignments that are not group assignments, copying or sharing assignments or exam information with other students except in group assignments, submitting as your own information from current or former students of this course, copying information from anywhere on the web and submitting it as your own work, and submitting anything as your own work which you have not personally created for this course. If you do wish to use materials that are not your own, please check with me (your instructor) ahead of time and cite your source clearly. When in doubt, ask first! .

Note: Course materials cannot be shared outside the course without the instructor's written permission.

The use of geeksforgeeks, stackoverflow or any other online help, including reviewing YouTube videos, to solve homework or exam problems is not allowed! You may use them only for guidance and or informational purposes, however citation of all consulted resources, outside of the course materials, must be provided with the assignment submission.

All code/work submitted by a student **must be developed wholly by the student**. Directly copying an alternative resource without crediting the source is considered plagiarism. You are allowed to collaborate on in-class Lab exercises, discuss approaches to assignments at a high level with each other, but you may not jointly develop homework or exam problem answers.

Generative AI Usage (eg, ChatGPT)

Gaining fluency in any programming or computer systems concepts takes sustained practice, trial and error, and the willingness to work through mistakes. This discomfort is normal and an essential part of learning.

With the recent prominence of Generative AI, it is expected that students become familiar with these tools in order to enhance productivity, and be prepared for their potential uses in future careers. At the same time, these tools should not impair the learning process, and prevent student retention of the material.

Data structures and algorithms are the building blocks of computer science and have a far-reaching impact on various fields. They are core to modern computing and are an absolute critical skill set in various fields, including software development, artificial intelligence and machine learning, web development, computer networking, gaming, cryptography, and more.

I do not recommend using generative AI for programming assignments. If a student wishes to use generative AI to obtain hints for programming assignments, a citation of the generative AI tool is required, with the assignment being marked down accordingly. In such cases, students are expected to critically evaluate the output of Generative AI, as the accuracy of answers is not guaranteed. All programming assignments require you to document your code to professional standards with comments (see 'Writing clear code' under useful cheat sheets for class on Sakai). Note: Students that rely heavily on Generative AI as a means to complete assignments will probably

find successful completion of the in-class assessments to be difficult, and risk not passing the course. More importantly, when searching for employment in any mid-size to larger company, it is 100% algorithm interviewing questions. Good Interviewers will spot candidates who do not know these core and fundamental topics within the first minute of questioning.

Acceptable use:

1. Explain a concept, or present an example for the purposes of clarifying a concept.
2. Explain code/examples presented in lecture, but not associated with programming assignments or homeworks.

Unacceptable use:

1. Using Generative AI to provide solutions to assignments in any part (whole or partially)
2. Using Generative AI to solve a "similar" problem to an assignment.
3. Using Generative AI to debug code associated with assignments

Privacy Statement

Assuring privacy among faculty and students engaged in online and face-to-face instructional activities helps promote open and robust conversations and mitigates concerns that comments made within the context of the class will be shared beyond the classroom. As such, recordings of instructional activities occurring in online or face-to-face classes may be used solely for internal class purposes by the faculty member and students registered for the course, and only during the period in which the course is offered. Students will be informed of such recordings by a statement in the syllabus for the course in which they will be recorded. Instructors who wish to make subsequent use of recordings that include student activity may do so only with informed written consent of the students involved or if all student activity is removed from the recording. Recordings including student activity that have been initiated by the instructor may be retained by the instructor only for individual use. Miscellaneous notes: This syllabus represents the status of the course and may be changed.

Language about Pronouns

Faculty at Loyola strive to foster inclusive classroom environments, both in-person and online. Respecting pronouns and using gender-inclusive language not only helps students feel like they belong in your classroom, but also better ensures that they are fully present in their academic experience and do not experience impediments to their learning. Developed by colleagues in the School of Law, the language below can set a tone of inclusivity from the start of any course. Use of Appropriate Names and Pronouns: Addressing one another at all times by using one's chosen modes of address (including preferred names and gender pronouns) honors and affirms individuals of all gender identities and gender expressions. Misgendering and heteronormative language excludes the experiences of individuals whose identities may not fit within a gender binary, and/or who may not

identify with the sex they were assigned at birth. If you wish, please share your gender pronouns with me and the class when you introduce yourself, on your name placard, and/or on your Zoom profile. If you do not wish to be called by the name that appears on the class roster or attendance sheet, please let me know privately and I will work diligently to honor your wishes. My goal is to create an affirming environment for all students so that everyone can learn and engage as our full and true selves.